Exercise 2: Worked solutions

Electrical Engineering I · Module 92315 · Exercise 2

From Fundamentals to AI Apocalypse · personal study guide

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Try each part before opening the answer. The steps build fields and forces from direction and distance, then check against symmetry, units, or a limiting case.

Task 4: Electric force compared with gravity

Protons at A carry $Q_{\mathrm{A}} = +1$ C and electrons at B carry $Q_{\mathrm{B}} = -1$ C; they are 1 km apart. Find $|F_{\mathrm{E}}|/|F_{\mathrm{G}}|$. Use $\gamma = 6.667 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $e = 1.602 \times 10^{-19}$ C, $m_{\mathrm{p}} = 1.672 \times 10^{-27}$ kg and $m_{\mathrm{e}} = 9.1094 \times 10^{-31}$ kg.

Connected topic: Coulomb's law (§1.2) gives the electric force; compare it with the supplied gravitational law.

4: Magnitude ratio

Try first: convert each 1 C charge quantity into a number of elementary charges, then form both force magnitudes before dividing.

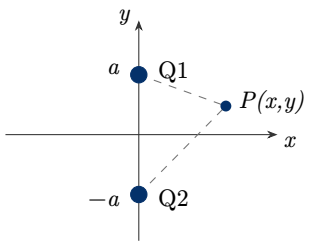
Answers: see "Task 4: Electric force compared with gravity" under Answers at the end.

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Task 5: Field from two charges on the y-axis

Place Q_1 at $(0,\ a)$ and Q_2 at $(0,\ -a)$. Find the field throughout the xy-plane and simplify it on the x-axis for the stated charge choices. With charges in coulombs and lengths in metres, field values are in N/C.

Connected topics: Electric field strength (§1.3) defines one source field; superposition (§1.4) adds them.



Resolve each contribution from its own source. P is arbitrary; its displacement from each charge sets direction and distance.

5a: Field anywhere in the plane

Try first: write the displacement from each source to P $= (x,\ y)$, then add both vectors.

Answers: see "Task 5: Field from two charges on the y-axis" under Answers at the end.

5b: Equal charges, $Q_1 = Q_2 = Q$

Try first: set $y = 0$ and identify which components cancel by symmetry.

Answers: see "Task 5: Field from two charges on the y-axis" under Answers at the end.

5c: Opposite charges, $Q_1 = Q$ and $Q_2 = -Q$

Try first: again set $y = 0$. Opposite signs change which components reinforce.

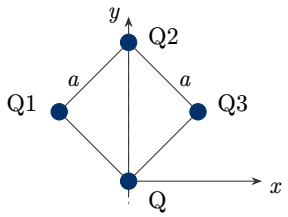
Answers: see "Task 5: Field from two charges on the y-axis" under Answers at the end.

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Task 6: Charges at the corners of a square

Four nonzero point charges occupy corners of a square of side a. Q is at the origin, Q_2 lies on the positive y-axis, and the surrounding permittivity is ϵ_0. With SI values, force is measured in newtons and work in joules.

Connected topics: Coulomb's law (§1.2) gives each force; superposition (§1.4) combines them. Work connects to electrostatic potential (§1.8).



The square is rotated in the coordinate plane. Sloping sides are a; the vertical diagonal Q to Q_2 is $\sqrt{2}a$.

6a: Force on Q at the origin

Try first: find the source coordinates from the diagram, then resolve each force into x and y components.

Answers: see "Task 6: Charges at the corners of a square" under Answers at the end.

6b: Condition for zero force

Try first: set both components from 6a to zero. The central charge Q is nonzero.

Answers: see "Task 6: Charges at the corners of a square" under Answers at the end.

6c: Work to move Q_2 to (0, a)

Try first: with $Q_1 = Q_3 = 0$, compare its initial and final distance from Q. Take work as external work for a quasistatic move.

Answers: see "Task 6: Charges at the corners of a square" under Answers at the end.

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Answers

Task 4: Electric force compared with gravity

4 · Coulomb force versus gravity

1. Gravity needs group masses, so first convert total charge to particle count. Each particle has charge magnitude e:

$$N = |Q|/e; \quad m_{\mathrm{A}} = |Q_{\mathrm{A}}|m_{\mathrm{p}}/e; \quad m_{\mathrm{B}} = |Q_{\mathrm{B}}|m_{\mathrm{e}}/e$$

2. Coulomb's law gives electric magnitude; opposite signs make the force attractive:

$$|F_{\mathrm{E}}| = k|Q_{\mathrm{A}}Q_{\mathrm{B}}|/r_{\mathrm{AB}}^2, \quad k = 1/(4\pi\varepsilon_0)$$

3. Insert the group masses into the given gravity relation:

$$|F_{\mathrm{G}}| = \gamma m_{\mathrm{A}} m_{\mathrm{B}}/r_{\mathrm{AB}}^2 = \gamma|Q_{\mathrm{A}}Q_{\mathrm{B}}|m_{\mathrm{p}}m_{\mathrm{e}}/(e^2 r_{\mathrm{AB}}^2)$$

4. Divide the magnitudes; the charge quantities and separation cancel:

$$|F_{\mathrm{E}}|/|F_{\mathrm{G}}| = k\,e^2/(\gamma m_{\mathrm{p}} m_{\mathrm{e}})$$

5. Substitute $k = 8.9877 \times 10^9$ N m^2/C^2 and the given constants:

$$|F_{\mathrm{E}}|/|F_{\mathrm{G}}| \approx 2.27 \times 10^{39}$$

Result: electric force is about 2.27×10^{39} times stronger in magnitude.

Check: the ratio is dimensionless and independent of distance since both laws vary as $1/r^2$.

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Task 5: Field from two charges on the y-axis

5a · General field by superposition

1. Point each displacement from source to P by subtracting source coordinates from point coordinates:

$$r_1 = x\,e_x + (y - a)e_y; \quad r_2 = x\,e_x + (y + a)e_y$$

2. A point-charge vector field is kQr/r^3; signed Q handles direction. Superpose the two source fields:

$$E(\mathrm{P}) = kQ_1 r_1/|r_1|^3 + kQ_2 r_2/|r_2|^3$$

3. Take the lengths from the displacement components and substitute:

$$E(x,y) = kQ_1[x\,e_x + (y - a)e_y]/[x^2 + (y - a)^2]^{3/2} + kQ_2[x\,e_x + (y + a)e_y]/[x^2 + (y + a)^2]^{3/2}$$

4. **Result:** valid except at $(0, \pm a)$, where an ideal point-charge field is singular.

   Check: a positive source points along its source-to-P displacement; a negative source reverses it.

Task 5: Field from two charges on the y-axis

5b · Equal charges on the x-axis

1. At $\mathrm{P} = (x, 0)$, the displacements have equal lengths and opposite vertical components:

$$r_1 = x\,e_x - a\,e_y; \quad r_2 = x\,e_x + a\,e_y; \quad |r_1| = |r_2| = \sqrt{(x^2 + a^2)}$$

2. With equal Q, vertical components cancel and horizontal components add:

$$E(x,0) = kQ(r_1 + r_2)/(x^2 + a^2)^{3/2} = 2kQxe_x/(x^2 + a^2)^{3/2}$$

3. **Result:** for positive Q it points +x when $x > 0$ and reverses for $x < 0$; negative Q reverses these directions.

   Check: at $x = 0$ symmetry requires zero, which the expression gives.

Task 5: Field from two charges on the y-axis

5c · Opposite charges on the x-axis

1. The distances still match, but signed contributions use the difference of the displacement vectors:

$$r_1 - r_2 = -2a\, e_y$$

2. Horizontal parts cancel; vertical contributions add downward:

$$E(x,0) = kQ(r_1 - r_2)/(x^2 + a^2)^{3/2} = -2kQae_y/(x^2 + a^2)^{3/2}$$

3. **Result:** for positive Q it points from upper positive to lower negative: −y. Negative Q reverses it.

   Check: at the midpoint both fields point down. Far along the axis, magnitude falls as $1/|x|^3$, as for a dipole.

Task 6: Charges at the corners of a square

6a · Vector sum at the origin

1. The square's 45° sides project equally on both axes. The other corners and their distances from Q are:

$$Q_1 : (-a/\sqrt{2},\, a/\sqrt{2}); \quad Q_2 : (0,\, \sqrt{2}a); \quad Q_3 : (a/\sqrt{2},\, a/\sqrt{2})$$

   Side charges are distance a; Q_2 is distance $\sqrt{2}$a.

2. The top charge contributes along −y for positive QQ_2; its distance is $\sqrt{2}$a:

$$F_2 = -kQQ_2/(2a^2)e_y$$

3. The side-charge unit directions from each source toward the origin give:

$$F_1 = kQQ_1/(\sqrt{2}a^2)(e_x - e_y); \quad F_3 = -kQQ_3/(\sqrt{2}a^2)(e_x + e_y)$$

4. Add components and factor the scale, where $k = 1/(4\pi\epsilon_0)$:

$$F = kQ/a^2\,[(Q_1 - Q_3)/\sqrt{2}\, e_x - ((Q_1 + Q_3)/\sqrt{2} + Q_2/2)e_y]$$

5. **Result:** this vector is the net force on Q; charge signs select attraction or repulsion.

   Check: each term has charge product over distance squared. If $Q_1 = Q_3$, symmetry removes the x-component.

Task 6: Charges at the corners of a square

6b · Cancel both components

1. Since $Q \neq 0$, the shared factor kQ/a^2 cannot make the force zero. The x-component requires:

$$(Q_1 - Q_3)/\sqrt{2} = 0 \quad \Longrightarrow \quad Q_1 = Q_3$$

2. The y-component must also vanish:

$$(Q_1 + Q_3)/\sqrt{2} + Q_2/2 = 0$$

3. Use $Q_3 = Q_1$ and solve:

$$Q_2 = -\sqrt{2}(Q_1 + Q_3) = -2\sqrt{2}Q_1$$

4. **Result:** $Q_1 = Q_3$ and $Q_2 = -2\sqrt{2}Q_1$.

   Check: equal side charges cancel horizontally; the top charge must have the opposite sign so its vertical force balances theirs.

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Task 6: Charges at the corners of a square

6c · Change in electrostatic potential energy

1. The diagram places Q_2 at $(0, \sqrt{2}a)$; the requested endpoint is $(0, a)$. Only the interaction with Q remains. This starting position is read from the diagram.
2. For two point charges, $U = kQQ_2/r$. Quasistatic external work equals the potential-energy change:

$$W_{ext} = \Delta U = kQQ_2(1/a - 1/(\sqrt{2}a))$$

3. Use $k = 1/(4\pi\epsilon_0)$:

$$W_{ext} = QQ_2/(4\pi\varepsilon_0 a)(1 - 1/\sqrt{2})$$

4. **Result:** the signed external work is positive for $QQ_2 > 0$ and negative for $QQ_2 < 0$.

   Check: the final separation is smaller. Like charges need positive external work to move closer; unlike charges release energy.

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